

Wang Ming

Gender: Male Objective: Senior Backend Engineer Location: Nanjing

Phone: 15834735977 Email: wangming6257@qq.com GitHub: <https://github.com/wangming515>

Education

2010.09 - 2014.06 Tongji University Artificial Intelligence (B.S.)

Courses: Computer Organization, Operating Systems, Computer Networks, Compilers, Database Systems, Design Patterns

Summary

Seven years of backend experience focused on payment/transaction systems, distributed architecture, high concurrency and production incident response.

Work Experience

2017.07 - Present Baidu Senior Backend Engineer

- Designed a Kafka-based async transaction pipeline with idempotency keys, retry queues and dead-letter handling, sustaining 20000 QPS at peak, significantly lowering maintenance cost.
- Investigated rising instance memory rss using GC logs, heap dump and arthas, found an unbounded cache and cut memory by 53%, ensuring online service stability.
- Owned backend of core transaction and order services, splitting the monolith into order, settlement and risk-control microservices.
- Served 1910K requests per day while keeping core-path SLA above 99.9%.
- Optimized a file-sort slow SQL on a hot API, added a composite index via gh-ost online DDL, reducing P99 from 1200ms to 220ms, significantly lowering maintenance cost.

2014.07 - 2017.06 Bilibili Backend Engineer

- Tuned Redis caching for hot keys and cache breakdown with local cache and distributed locks, raising hit rate above 90%, significantly lowering maintenance cost.
- Built service observability covering API latency, error rate, JVM GC, thread pools and consumer backlog with Prometheus and Grafana, significantly lowering maintenance cost.
- Rebuilt the config and template management service to support hot template reload without restart, cutting release time by 41%, improving end-to-end delivery quality.
- Led a production incident review with root-cause analysis and drove an alert-tiering and on-call rotation mechanism, supporting rapid business growth.

Projects

Distributed Payment & Settlement Platform (Spring Boot + MySQL + Kafka + Redis)

- Recorded an audit trace id for every payment state transition for compliance.
- Used sharding and read-write splitting to keep single-table rows within tens of millions.
- Implemented daily reconciliation across payment gateway, ledger and settlement files with auto-correction, improving end-to-end delivery quality.
- Applied idempotency and eventual consistency to avoid duplicate charges on retry, significantly lowering maintenance cost.

Enterprise Auth & Gateway Platform (Spring Cloud Gateway + OAuth2)

- Integrated OAuth2 and RBAC with multi-tenant isolation, recognized by stakeholders and peers.
- Unified auth, rate limiting, gray routing and tracing as reusable platform capabilities.
- Shipped a unified SDK to lower integration cost across teams.
- Added circuit breaking to isolate faulty downstreams, significantly lowering maintenance cost.

High-Concurrency Flash-Sale & Inventory System (Java + Redis + RocketMQ)

- Pre-deducted inventory in Redis with async DB writes to absorb traffic spikes, and key metrics improved steadily.
- Built circuit breaking and fallback for downstream failures, and key metrics improved steadily.
- Stayed stable at 20000 QPS in load tests with error rate below 0.1%, and key metrics improved steadily.
- Designed distributed locks and token-bucket rate limiting to prevent oversell, recognized by stakeholders and peers.

Highlights

- Continuously optimized project management and planning, cutting key-path latency from 1200ms to 220ms
- Owned core modules that steadily served 1910K daily visits with 99.9%+ availability
- Built a reusable methodology and docs around MySQL indexing and slow-query tuning, adopted by several teams
- Established metrics and reviews around Redis caching and distributed locks to keep improving delivery quality
- Drove a focused effort on CI/CD pipelines, reducing related issues by about 41%

Skills

Core skills: Microservices, JVM, MySQL indexing and slow-query tuning, Spring Boot, Observability

Proficient in: Java, Redis caching and distributed locks, Kafka/RocketMQ messaging, High Concurrency, JVM tuning and troubleshooting

Familiar with: SQL Optimization, Docker and containerization, MySQL, Kafka, Redis, Distributed transactions and consistency

Self Evaluation

- Stays calm under pressure and recovers services quickly during high-concurrency incidents
- Curious about new tech, continuously learning LLM and cloud-native topics and applying them in practice